

The Project, Complete

Findings, sealed evidence, open questions and the asks — 22 August 2026

Dheirav

August 24, 2026

The project, complete — 25 August 2026

This is the whole project in a few pages, written so it can be read instead of the paper. The paper (15 pages, IEEE format, before the planned compression pass that waits on your page-limit answer) exists and is current; everything here points at a result that is committed in the repository, and every number below is re-derived from stored results by an automated check before it is allowed to appear in a document. Since the previous version of this report: the paper has been through three adversarial review panels (twenty-four independent reviews in total, no reject verdicts), every cheap fix they converged on is applied, an apparent 2022 anomaly has been traced to its mechanism and resolved, and an overnight self-audit then re-tested our own numbers — killing one more hypothesis, downgrading one published number, and closing one open question, all recorded below.

The question, and why it matters

A fairness constraint asks a model to give two groups similar approval rates, but it does not say how the gap should close. It can close because more people from the disadvantaged group are approved, or because people from the advantaged group lose approvals they would have had. Both count as success, and the fairness metric — the number a company would report and a regulator would check — comes out the same in both cases. This means a team imposing a constraint cannot tell, from its own dashboard, whether it is about to extend opportunities or withdraw them.

The literature already knew the metric was silent about this. What it did not have is a way to know **in advance** which of the two outcomes a given system will produce. That is what this project measured, and the answer turns out to be readable from one number every deployed system already has: the rate at which its current model says yes.

The finding

Below a crossover in the baseline approval rate — empirically near 0.54 — an attribute-blind parity constraint shrinks the total pool of favourable decisions, while above it the same constraint grows the pool. On UCI Adult (rate 0.24), six different mitigation methods all shrink the pool, by 7.9% to 22.1%. On mortgage approvals (rate 0.81), the identical constraint grows it by 4.3%. The parity metric reports success identically in both cases, which is exactly the problem: the direction is invisible to the certificate but predictable from the rate.

The claim is deliberately narrow, and each boundary is measured rather than assumed. It holds within a population, in the attribute-blind in-processing regime, and it predicts the direction only — magnitude is ordered within a population (Spearman rho up to +0.96) but does not transfer across populations, where a pooled slope reaches only +0.487 against a pre-registered bar of +0.70 and therefore fails.

Why the evidence is stronger than a correlation

Every correlation in the project was computed after the arms were run, which is the weakest support for a claim of the form "you can know in advance". So the rule was tested the only way that format can be tested: **sealed predictions, committed to git before the data existed**.

The first sealed test failed, and the failure was informative. A refinement added from four in-sample populations two hours before the seal turned what would have been 7 of 8 into **4 of 8**, no better than a constant. The refinement was withdrawn, and the simple rule — down below 0.54, up at or above — was re-sealed against ten states the project had never touched, with the bar (at least 9 of 10, and strictly beating the best constant) committed before any run.

It scored 9 of 10, against a best constant of 6. The one miss is charged against the rule under the criterion sealed with it, and it is an instructive miss: an arm whose true effect is around -0.04% and flips sign between random seeds, which means there may have been no direction there to predict.

The complete ledger of registered tests is below, and one thing matters for reading it: these are not seven attempts at one claim, of which two succeeded. Each test targeted a **different** claim, and each failure removed a claim *stronger* than the one that survives — so the failures are boundary measurements, and the passes test the claim as those boundaries define it. Averaging the rows into "2 out of 7" would be reading a map as a scorecard.

Test	Outcome	What it settled
HMDA held-out sweep	2 of 4 — fail	our measuring procedure is unreliable on mortgage data, so no claim rests on it there
Equalized-odds transfer	+0.334 vs +0.70 — fail	the rule works for parity, which controls approval rates directly; it failed for equalized odds, which does not
Post-processing transfer	$r = -0.024$ — fail	the rule stops working the moment the model may read the protected attribute — and that turned out to be the boundary the theory predicts
Selection-rate derivation	beaten by a constant — fail	we cannot explain why the rule works; the paper claims only that it does
Sealed rule, refined	4 of 8 — fail	the extra clause we had added from early data was wrong, and is withdrawn
Attribute-aware consistency	9 of 9 — holds	where the attribute is readable, the disadvantaged group always gains; a consistency check with the theorem, which had already made this outcome near-certain — counted as confirmation, not as forecasting skill
Re-seal, unrefined rule	9 of 10, constant 6 — holds	the simple rule predicts populations it has never seen, from the approval rate alone
Crossover-residual test	3 located vs floor 6 — under-powered	what predicts the switch point's <i>location</i> stays unjudged; a third of large states have no switch point to locate
Sealed magnitude model	MAE 4.50 vs 0.77 — fail	the effect's <i>size</i> cannot be predicted from what we tried; the direction-only concession is earned
Sealed shape boundary	4 of 6 — fail	the base-rate boundary called every 2014 shape and both predicted flips; the 2022 curves misbehaved for a reason found later (below)
Sealed attribute-independence	2 of 6 — fail	a population's curve shape is not a property of its label alone; the race arms behave differently from the sex arms
Sealed cross-task shape	0 of 4 — fail	on two brand-new prediction tasks every scored call inverted: the shape boundary is a property of the income question itself, and only there

Nine of the twelve rows are failures or refusals, reported uncorrected, which removes the usual way lucky results reach print: selective reporting. The passes carry their own health warnings, stated in the paper where the claims are made. The one-in-twenty chance figure for the 9 of 10 is against an independent guesser; the stricter paired comparison — did the rule beat the best constant on the same ten arms — comes down to the five calls they disagree on, of which the rule wins four, about one in five by chance. And the sealed cohort has a design confound the paper states next to

the score: every arm reaches its approval rate through the income cutoff, so a rule reading only the cutoff would score identically there. The experiment that separates them is designed, committed, and waiting on data (the census extract below).

The relationship to the 2026 theory paper

Fairness May Backfire (arXiv:2603.06901) proves that in the attribute-blind regime the direction is distribution-dependent, so the conditionality itself is theirs and the paper says so. What they do not have is any way to compute the direction: their conditions are stated over a quantity that diverges on real data, and on 26 real arms from 15 populations they can be evaluated on none. Their paper contains no experiments and no occurrence of "selection rate" or "base rate", which was checked against the full text.

The project's position relative to them is two-sided, and both sides are measured. Their regime distinction is right: under attribute-aware post-processing the selection-rate relationship vanishes entirely ($r = -0.024$ against $+0.585$ on the same populations), and their prediction for that regime — the disadvantaged group always gains — holds on **27 of 27 populations, nine of them pre-registered**. And their structural quantity is tracked by the plain selection rate at $r = +0.935$, which means the number this project uses is a computable proxy for the quantity their theorem is stated over. They proved the question has an answer; this project built the instrument that reads it.

Who actually pays, which the metric also hides

Measured in rates, the burden of a fairness fix looks nearly even. Measured in people it is not: on Adult, between 66% and 74% of the individuals whose decision changed were advantaged-group members losing an approval, at 2.68 approvals destroyed per one created. Adding a one-line floor to the objective — "do not shrink the pool" — drops that exchange rate to 0.88 across nineteen arms at a cost of about 0.12 accuracy points, so the harm is optional once you know to ask for it. Knowing when to ask is what the crossover rule is for.

A 2022 anomaly, diagnosed to its mechanism and resolved

Four 2022 state-years initially appeared to *invert* the relationship — the opposite pattern to every earlier vintage — and for two days that stood as the project's most worrying open item. Three candidate mechanisms were then run to ground. Pandemic-era survey weighting: exculpated (applying the Census Bureau's person weights moves 2022's rates and gaps no more than 2018's). A recoded survey column: acquitted (2019 files carry the recode and behave classically). The third candidate is confirmed as the substantial cause: the \$50,000 income label, fixed in nominal terms, had slid down an inflation-compressed earnings distribution — the task had changed under the label. Re-measured at \$60,000, the 2022 real-equivalent of 2018's \$50,000, Ohio returns to a clean crossing at its old location and the inverted patterns disappear. Measured at constant real value there is no detected change in the phenomenon from 2014 through 2022. The practical lesson made it into the paper's abstract: a fixed nominal outcome definition is a different task each year, so labels should be anchored in real terms and teams should audit their own current data.

Three rounds of adversarial review, and what they changed

The paper was put through three review panels of independent reviewers with instructions to be harsh: six, then eight, then ten reviewers, the last round spanning lenses from a bank examiner and an EU regulator to a hostile senior skeptic told to argue the work is trivial. Twenty-four reviews, no reject verdicts; the skeptic conceded the trivial-arithmetic reading fails (an effect that vanishes

under post-processing cannot be rate arithmetic). Every converged fix is applied. The ones that produced new results:

- **The audit's own totals.** Run over all 45 swept population-label pairs, the procedure returns a usable direction on 27 (never contradicting what the constraint actually did), declares no-crossover on 6, and refuses on the rest — it answers six times in ten, with the refusals counted rather than hidden.
- **The "lottery" is bounded.** The earlier finding that a constraint can close the gap by discarding approvals at random, under an unchanged certificate, was probed at normal operating points as a control: it does not appear there in any of seven populations. It is a severe-operating-point phenomenon — a checkable red flag, not the ordinary cost of fairness.
- **The deployable form preserves the verdict.** A regulated lender cannot deploy a randomized model, so the deterministic extraction was tested directly: it agrees with the randomized version's direction on 10 of 10 populations, with the fairness repair intact. The examiner's objection is answered with data.
- **The seeds tell the miss's story.** 16 of the 19 sealed arms are unanimous across random seeds; the three that split are exactly the near-zero effects — and the one sealed miss (Minnesota, -0.04%) splits 3-of-5 *identically* to an arm that scored correct ($+0.04\%$). The audit now refuses such arms as indeterminate instead of scoring them.
- **Verifiability.** All eleven registered tests now have their registration and scoring commits pinned in the paper (each ordering re-verified), a checksum manifest pins the exact input files, and the paper states plainly that the chronology currently rests on our own clock — the next sealed test will be anchored with an external timestamp.

The overnight self-audit, and what it subtracted

After the review rounds, four of our own numbers were put under load in one sealed and three post-hoc tests (24 Aug):

- **A twelfth registered test, and a clean kill.** The curve-shape boundary — base rate below 0.365 means one family, above means the other — was sealed against six arms of two prediction tasks this project had never run, with an external (Bitcoin-anchored) timestamp on the seal itself. It scored **0 of 4, every call inverted**: after failing across years and across attributes, it has now failed across tasks, so the boundary belongs to the income question specifically. Meanwhile the *direction* rule's expectation held at all six new arms — including one where the advantaged group inverts — so the central claim came through untouched.
- **One published number downgraded.** Re-testing the four located crossovers under a nested bootstrap (refitted models $\times$ resampled test rows), three survive — but the Dutch census's 0.576 never brackets a crossing at all, in 1,000 of 1,000 resamples. It was a property of the six arms that produced it, and the paper now says so.
- **One open question closed, in the softer direction.** Whether the "lottery" is an optimiser mistake a vendor could avoid: no — at those operating points *no* score-informed blind alternative of the same size exists at all. The wrong is exactly and only that the lottery happens unannounced under a clean certificate.
- **The lift-versus-cut mechanism made visible.** A decile-by-group table shows the optimiser either grades probability into the disadvantaged group's mid-scores, or grants nothing below the top decile and cuts by lottery — the open selector question now has a concrete object to explain.

Scale, and the discipline behind the numbers

The record holds **57 independent populations** across seven data sources (UCI Adult, ACS states, HMDA mortgages, COMPAS, LSAC, the Dutch census, Taiwanese credit-card default), six decision domains, three countries and two decades. The later campaigns mostly went against us, in instructive ways: the test of what predicts the crossover's location returned **underpowered** (three located against a floor of six), a sealed model of the effect's *size* lost outright to predicting zero, and the three crossovers that did locate **broke the cluster** — Florida at 0.284 and Pennsylvania at 0.652, far outside the 0.43–0.58 band, on the same instrument that built it. Several of the largest states turn out to have no crossover at all, and the fixed 0.54 prior scores 5 of 10 on their natural arms, each miss agreeing with that state's own measured crossover. So the within-population claim survived the campaign's hardest tests, while the transferable prior narrowed to "usually, not always" — and every one of those sentences comes from a prediction committed before the data existed.

Three pieces of discipline are worth naming because they caught real errors:

- **55 automated checks** re-derive every documented figure from stored results; one of them caught the paper's population count going stale within minutes of new results landing, more than once.
- **Population counts are recomputed, never quoted** — three counts in the project's own history turned out to be arm counts masquerading as population counts, and the paper now carries a table saying exactly which populations enter which analysis.
- **Anything that could go either way is sealed first.** The standing rule commits predictions, thresholds and the naive baseline to beat before the run; re-analysis of existing results is labelled post-hoc every time it appears.

What is open, honestly

- **Why the crossover sits where it sits.** Located values span 0.28–0.65 and nothing predicts the location; the sealed test built to answer this returned underpowered, a third of the large states have no crossover to locate, and the nested bootstrap just reduced the solidly located non-lending crossovers from four to three. What governs a population's curve *shape* is the sharpest form of this question, and we now know three things it is **not**: the survey year (that was the label sliding), the attribute alone, and — as of the twelfth registered test — the label's base rate independent of the task.
- **Breadth.** The sealed 9-of-10 is ten US census populations — depth, not breadth. A non-Western census extract (Brazil 2000/2010, Mexico 2015/2020) is requested and pending approval, which would allow the first crossover located outside the West — and it carries the one experiment the reviews agree matters most: the sealed cohort that separates the approval rate from the income cutoff, its design already committed.
- **Mechanism.** At extreme operating points the constraint closes the gap by two qualitatively different solutions — lifting the disadvantaged versus discarding approvals — and what selects between them is measured to be none of six candidate quantities. The paper states this as an open boundary rather than papering over it.

What I am asking

1. **Venue and page limit.** The paper is 15 pages in IEEE conference format after the review rounds' additions; a compression pass with its cut list already written brings it to roughly

9-10 without losing any number, and your venue answer (FAccT/AIES ~10, IEEE conferences ~8) decides how deep it goes.

2. **Does the empirical half stand as a paper, given the theory paper exists?** My position: their theorem is uncomputable on data and this project supplies the measurement, the boundary verification and a sealed predictive rule — but this is the call I need from you.
3. **AI disclosure.** How the department wants assistance disclosed, including whether commits should carry a trailer; none currently do.
4. **Whether the course submission window is still open** for the rebuilt report and deck.
5. **Contacting the theory paper’s authors.** Our results confirm their regime prediction (27 of 27, nine pre-registered), extend their framework with a computable proxy for its central quantity, and show its stated conditions cannot be evaluated on real data — the last of which rests on an estimator choice they would be the best people to challenge before a reviewer does. Standard practice would be a short collegial email, ideally after a preprint exists; I would like your read on whether and when, and whether you want to be on it.